Final Mock Exam — Set C — Solutions

Lectures 1–12 (comprehensive)

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Overview of the final mock exam — Set C

Problem	Topic	Lecture slides	Points
P1	Moving beyond linearity — splines	Ch. 7 — Moving Beyond Linearity (Lecture 9)	16
P2	Generalised additive models	Ch. 7 — Moving Beyond Linearity (Lecture 9)	8
P3	Trees by hand (regression & classification)	Ch. 8 — Tree-Based Methods (Lecture 10)	20
P4	Ensembles — bagging, RF, boosting	Ch. 8 — Tree-Based Methods (Lecture 10)	16
P5	Neural networks by hand	Ch. 10 — Deep Learning (Lecture 11)	20
P6	Multiple testing	Ch. 13 — Multiple Testing (Lecture 12)	20
P7	Cumulative essentials	Ch. 2–6 (Lectures 1–8)	20
Σ			**120**

Exam conditions

120 minutes for 120 points ($\approx$ 1 point per minute). Allowed aids: non-programmable calculator and one handwritten A4 sheet (both sides). All required constants (e-powers and logs) are given in the problems. Justify every answer.

Problem 1 (16 P) — Moving beyond linearity

Lecture slides: Chapter 7 — Moving Beyond Linearity (Lecture 9).

(a) **[4 P]** How many degrees of freedom does a cubic spline with K interior knots use? Justify by counting the parameters of the pieces and subtracting the knot constraints; evaluate for $K = 3$.

(b) **[4 P]** Degrees of freedom of a *natural* cubic spline with the same $K = 3$ knots; which extra constraint acts where, and why is it useful?

(c) **[4 P]** Smoothing spline $\sum_i (y_i - g(x_i))^2 + \lambda \int g''(t)^2 dt$: role of the two terms; behaviour of $\hat{g}$ as $\lambda \to 0$ and $\lambda \to \infty$.

(d) **[4 P]** Truncated-power basis of a cubic spline with one knot ξ; define $(x - \xi)_+^3$; why does adding it keep the fit smooth at ξ?

Problem 1 — solution (a): df of a cubic spline

Solution P1(a) — degrees of freedom of a cubic spline

Step 1 — count the free parameters. K interior knots split the range of X into $K+1$ sub-intervals; on each the fit is a *separate* cubic with 4 coefficients:

$$\underbrace{(K+1)}_{\text{pieces}} \times \underbrace{4}_{\text{coeffs/piece}} = 4(K+1) = 4K+4 \quad \text{parameters.}$$

Step 2 — subtract the knot constraints. At each knot the two adjacent cubics agree in value, first and second derivative (g, g', g'' continuous): 3 per knot, $3K$ in total.

$$\mathrm{df} = 4(K+1) - 3K = 4K + 4 - 3K = K + 4 \quad \Longrightarrow \quad K = 3: \ \mathrm{df} = 3 + 4 = 7.$$

Basis cross-check

Basis $\{1, x, x^2, x^3\}$ plus one truncated power per knot: $4 + K = 4 + 3 = 7$ — same count.

Problem 1 — solution (b): natural cubic spline

Solution P1(b) — df of a natural cubic spline

Extra constraint. A *natural* cubic spline is additionally required to be *linear beyond the two boundary knots*, i.e. in the outer regions $X < \xi_1$ and $X > \xi_K$.

Cost of the constraint. Linearity forces the quadratic *and* cubic terms to vanish in each outer region — 2 parameters removed per boundary region, $2 \times 2 = 4$ in total:

$$\mathrm{df} = (K + 4) - 4 = K \qquad \Longrightarrow \qquad K = 3: \ \mathrm{df} = 3.$$

Where it acts and why it is useful. The constraint bites only in the two tails, exactly where the data are sparse and ordinary splines/polynomials are most erratic. Forcing linearity there lowers the *variance* of predictions at and beyond the boundary knots, at the price of a little extra bias — a favourable trade near the edges.

Problem 1 — solution (c): smoothing spline

Solution P1(c) — roles of the two terms and the limits

$$\hat{g} = \arg\min_{g} \underbrace{\sum_{i=1}^{n}\big(y_i - g(x_i)\big)^2}_{\text{RSS: fidelity to the data}} + \lambda \underbrace{\int g''(t)^2\,dt}_{\text{roughness (curvature) penalty}} .$$

The RSS rewards fitting the points; $\int g''(t)^2 dt$ measures total curvature (wiggliness); $\lambda \geq 0$ tunes the trade-off.

- $\lambda \to 0$: the penalty disappears $\Rightarrow$ $\hat{g}$ can be arbitrarily wiggly and *interpolates* every training point (training RSS $= 0$, severe overfitting; effective df $\to n$).
- $\lambda \to \infty$: any curvature is infinitely expensive $\Rightarrow$ $g'' \equiv 0$ is enforced $\Rightarrow$ $\hat{g}$ is the *least-squares straight line* (effective df $\to 2$).

In between, λ (chosen e.g. by LOOCV) drives the effective df smoothly from n down to 2.

Problem 1 — solution (d): truncated-power basis

Solution P1(d) — one-knot cubic spline

Representation (single knot at ξ):

$$f(x) = \beta_0 + \beta_1 x + \beta_2 x^2 + \beta_3 x^3 + \beta_4 (x-\xi)_+^3, \qquad (x-\xi)_+^3 = \begin{cases} (x-\xi)^3, & x > \xi \\ 0, & x \le \xi. \end{cases}$$

Why the smoothness at ξ survives. Evaluate the added function and its derivatives at the knot (from the right, $x \downarrow \xi$):

$$(x-\xi)_+^3\big|_\xi = 0, \qquad 3(x-\xi)^2\big|_\xi = 0, \qquad 6(x-\xi)\big|_\xi = 0.$$

Value, first and second derivative are all 0 at ξ, so adding $\beta_4(x-\xi)_+^3$ leaves f, f', f'' continuous there; only the *third* derivative jumps (by $6\beta_4$). That is exactly the defining smoothness of a cubic spline.

Problem 2 — task

Problem 2 (8 P) — Generalised additive models

Lecture slides: Chapter 7 — Moving Beyond Linearity (Lecture 9).

(a) **[3 P]** General form of a GAM for quantitative Y; how does it generalise multiple linear regression?

(b) **[3 P]** One advantage concerning the interpretation of the f_j; the main structural limitation — and its repair.

(c) **[2 P]** Backfitting in one or two sentences.

Problem 2 — solution (a): GAM model form

Solution P2(a) — general form of a GAM

General form (quantitative response Y, predictors $X_1, \ldots, X_p$):

$$Y = \beta_0 + f_1(X_1) + f_2(X_2) + \cdots + f_p(X_p) + \varepsilon = \beta_0 + \sum_{j=1}^{p} f_j(X_j) + \varepsilon,$$

where each f_j is a smooth (possibly non-linear) function of a single predictor — e.g. a natural or smoothing spline, or a local regression.

How it generalises multiple linear regression. MLR is the special case in which every f_j is forced to be linear:

$$f_j(X_j) = \beta_j X_j \implies Y = \beta_0 + \beta_1 X_1 + \cdots + \beta_p X_p + \varepsilon.$$

A GAM relaxes this by letting each f_j be an arbitrary smooth curve, while keeping the *additive* structure (the predictor effects still simply add up).

Problem 2 — solution (b): advantage and limitation

Solution P2(b) — interpretability vs. no interactions

Advantage (interpretability). Because the model is additive, the contribution of each predictor is a single function f_j that can be *plotted and read on its own* (one panel per predictor), holding the others fixed. A non-linear effect is therefore as easy to communicate as a row in a coefficient table.

Structural limitation. The additive form allows *no interactions*: the shape of X_j's effect cannot depend on the value of X_k, so any genuinely synergistic effect is missed.

Repair. Add interaction components by hand — e.g. a product term $\beta_{jk}X_jX_k$ or a two-dimensional smooth surface $f_{jk}(X_j, X_k)$ — as an extra term in the model.

Problem 2 — solution (c): backfitting

Solution P2(c) — the backfitting algorithm

Idea. Fit the p functions *one at a time*, each time treating the others as known, and iterate.

One pass. Cycle through $j = 1, \ldots, p$ and re-fit f_j with a smoother (e.g. a spline) to the *partial residuals*

$$r_i^{(j)} = y_i - \beta_0 - \sum_{k \neq j} f_k(x_{ik}), \qquad i = 1, \ldots, n,$$

i.e. the response with every *other* current fitted effect removed.

Iterate the cycle until the fitted functions f_j stop changing (convergence). This sidesteps having to estimate all p smooth functions simultaneously.

Problem 3 (20 P) — Trees by hand

Lecture slides: Chapter 8 — Tree-Based Methods (Lecture 10).

Delivery vans x and parcels delivered y (1,000) for $n = 6$ depots; one split at s: $R_1 = \{x < s\}$ and $R_2 = \{x \geq s\}$ with region-mean predictions.

i	1	2	3	4	5	6
x_i (vans)	1	2	3	4	5	6
y_i (parcels)	3	5	4	9	11	10

(a) **[8 P]** Evaluate the cutpoints $s = 2.5$ and $s = 3.5$ by total RSS; which is chosen? (b) **[2 P]** Using the chosen split $s = 3.5$ from (a), predict for a new depot with $x = 5$ vans. (c) **[6 P]** Node with 5 "Electronics" / 3 "Clothing" / 1 "Books": proportions; Gini; entropy ($\ln \frac{5}{9} = -0.5878$; $\ln \frac{1}{3} = -1.0986$; $\ln \frac{1}{9} = -2.1972$); predicted class. (d) **[4 P]** Cost-complexity pruning: role of α in $\sum_m \sum_{i \in R_m} (y_i - \hat{y}_{R_m})^2 + \alpha |T|$.

Problem 3 — solution (a): cutpoint $s = 2.5$

Solution P3(a) part 1 — evaluate $s = 2.5$

Reference (root node). $\bar{y} = \frac{3+5+4+9+11+10}{6} = \frac{42}{6} = 7$; root $\mathrm{RSS} = \sum_i (y_i - 7)^2 = 16+4+9+4+16+9 = 58$.

Split at $s = 2.5$**:** $R_1 = \{x_i < 2.5\} = \{1, 2\}$, $R_2 = \{x_i \geq 2.5\} = \{3, 4, 5, 6\}$.

$$\hat{y}_{R_1} = \frac{3+5}{2} = 4, \qquad \hat{y}_{R_2} = \frac{4+9+11+10}{4} = \frac{34}{4} = 8.5.$$

$$\begin{aligned}
\mathrm{RSS}_{R_1} &= (3-4)^2 + (5-4)^2 = 1 + 1 = 2, \\
\mathrm{RSS}_{R_2} &= (4-8.5)^2 + (9-8.5)^2 + (11-8.5)^2 + (10-8.5)^2 \\
&= 20.25 + 0.25 + 6.25 + 2.25 = 29,
\end{aligned}$$

$$\implies \mathrm{RSS}(2.5) = 2 + 29 = 31.$$

Problem 3 — solution (a): cutpoint $s = 3.5$ and the choice

Solution P3(a) part 2 — evaluate $s = 3.5$ and decide

Split at $s = 3.5$: $R_1 = \{1, 2, 3\}$, $R_2 = \{4, 5, 6\}$.

$$\hat{y}_{R_1} = \frac{3+5+4}{3} = \frac{12}{3} = 4, \qquad \hat{y}_{R_2} = \frac{9+11+10}{3} = \frac{30}{3} = 10.$$

$$\mathrm{RSS}_{R_1} = (3-4)^2 + (5-4)^2 + (4-4)^2 = 2, \quad \mathrm{RSS}_{R_2} = (9-10)^2 + (11-10)^2 + (10-10)^2 = 2,$$
$$\implies \mathrm{RSS}(3.5) = 2 + 2 = 4.$$

s	$\hat{y}_{R_1}$	$\hat{y}_{R_2}$	total RSS
2.5	4	8.5	31
3.5	4	10	4

Recursive binary splitting is *greedy* and picks the smallest RSS: $s = 3.5$ ($4 \ll 31$; in fact the best of all five cutpoints), cleanly separating the low-volume depots (mean 4) from the high-volume depots (mean 10).

Problem 3 — solution (b): prediction for a new depot

Solution P3(b) — predict for $x = 5$ vans

Use the chosen split $s = 3.5$ and locate the new depot:

$$x = 5 \;\geq\; 3.5 \quad \Longrightarrow \quad \text{region } R_2 = \{x \geq 3.5\}.$$

The tree predicts the region mean of R_2:

$$\hat{y} = \hat{y}_{R_2} = 10 \quad \Longrightarrow \quad 10\,000 \text{ parcels.}$$

Reading a regression tree

A single-split regression tree is a step function: *every* $x \geq 3.5$ gets the same prediction $\hat{y}_{R_2} = 10$, regardless of how far above 3.5 it lies.

Problem 3 — solution (c): Gini and entropy

Solution P3(c) — impurity of a 3-class node

Counts: 5 Electronics, 3 Clothing, 1 Books ($n = 9$). Proportions:

$$\hat{p}_{\mathrm{E}} = \tfrac{5}{9}, \qquad \hat{p}_{\mathrm{C}} = \tfrac{3}{9} = \tfrac{1}{3}, \qquad \hat{p}_{\mathrm{B}} = \tfrac{1}{9}.$$

Gini $G = \sum_k \hat{p}_k(1 - \hat{p}_k)$:

$$G = \tfrac{5}{9} \cdot \tfrac{4}{9} + \tfrac{3}{9} \cdot \tfrac{6}{9} + \tfrac{1}{9} \cdot \tfrac{8}{9} = \tfrac{20+18+8}{81} = \tfrac{46}{81} = 0.568.$$

Entropy $D = -\sum_k \hat{p}_k \ln \hat{p}_k$ (using $\ln \tfrac{5}{9} = -0.5878$, $\ln \tfrac{1}{3} = -1.0986$, $\ln \tfrac{1}{9} = -2.1972$):

$$D = -\big(\tfrac{5}{9}(-0.5878) + \tfrac{1}{3}(-1.0986) + \tfrac{1}{9}(-2.1972)\big) = 0.3266 + 0.3662 + 0.2441 = 0.937.$$

Predicted class: the majority, **Electronics** ($\hat{p}_{\mathrm{E}} = 5/9 \approx 0.556$). Both G and D vanish for a pure node; here they are large because no class dominates.

Problem 3 — solution (d): cost-complexity pruning

Solution P3(d) — role of α

$$\min_{T \subseteq T_0} \underbrace{\sum_{m=1}^{|T|} \sum_{i:\, x_i \in R_m} (y_i - \hat{y}_{R_m})^2}_{\text{training RSS of } T} + \alpha \underbrace{|T|}_{\#\text{leaves}} .$$

$\alpha \geq 0$ is the *price per terminal node*, trading fit against complexity:

- $\alpha = 0$: the criterion is just the training RSS $\Rightarrow$ the full tree T_0 wins (nothing is pruned).
- $\alpha \uparrow$: each extra leaf must "pay for itself"; a branch is pruned once its RSS reduction falls below its cost $\Rightarrow$ a *nested sequence* of ever-smaller subtrees (weakest-link pruning).

Choosing α: by *cross-validation* — pick the α with the smallest CV error, then refit that subtree on all the data. This balances bias against variance rather than merely minimising training error.

Problem 4 (16 P) — Ensembles

Lecture slides: Chapter 8 — Tree-Based Methods (Lecture 10).

(a) **[6 P]** B identically distributed trees with variance σ^2 and pairwise correlation ρ: $\mathrm{Var}\left(\frac{1}{B}\sum_b \hat{f}_b(x)\right) = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$. (i) Why does this show variance reduction and what limits it? (ii) Evaluate for $B = 50$ and $\rho = 0.4$ and $\sigma^2 = 16$; compare with one tree. (iii) Limit as $B \to \infty$?

(b) **[4 P]** Random forests: modification at each split; recommended m for *regression* with $p = 30$; using $\mathrm{Var} = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$ from (a), why does it improve on bagging?

(c) **[4 P]** Boosting: from a call `GradientBoostingRegressor(n_estimators=5000, learning_rate=0.01, max_depth=1)`, what does each argument control? The slow-learning idea.

(d) **[2 P]** What does the OOB error estimate and why is it free?

Problem 4 — solution (a): variance of the bagged average

Solution P4(a) — reduction / evaluation / limit

$$\mathrm{Var}\left(\frac{1}{B}\sum_{b=1}^{B}\hat{f}_b(x)\right) = \underbrace{\rho\sigma^2}_{\text{correlated part (floor)}} + \underbrace{\frac{1-\rho}{B}\sigma^2}_{\to 0 \text{ as } B\to\infty} .$$

(i) Averaging shrinks only the second term (the trees' *independent* variability) to 0; the first term $\rho\sigma^2$ is free of B. Bootstrap trees share the same data and strong predictors, so $\rho > 0$ — this correlated part cannot be averaged away and *limits* the reduction.

(ii) For $B = 50$, $\rho = 0.4$, $\sigma^2 = 16$:

$$\mathrm{Var} = 0.4 \cdot 16 + \frac{1-0.4}{50} \cdot 16 = 6.4 + \frac{0.6\cdot 16}{50} = 6.4 + 0.192 = 6.592,$$

vs. $\sigma^2 = 16$ for a single tree — a reduction of $\frac{16-6.592}{16} \approx 58.8\%$.

(iii) $B \to \infty$: $\mathrm{Var} \to \rho\sigma^2 = 0.4 \cdot 16 = 6.4$ (the floor). More trees never overfit; they only stabilise the average.

Problem 4 — solution (b): random forests

Solution P4(b) — decorrelating the trees

Modification. At *each split*, only a fresh random subset of m of the p predictors is allowed as splitting candidates (bagging is the special case $m = p$).

Recommended m (regression): $m \approx p/3$, so

$$p = 30 \implies m = \tfrac{30}{3} = 10.$$

Why it beats bagging. With bagging a dominant predictor heads the top split of nearly every tree $\Rightarrow$ the trees look alike $\Rightarrow$ ρ is high. Withholding it from $\approx (p - m)/p$ of the split searches *decorrelates* the trees, i.e. lowers ρ. In $\mathrm{Var} = \rho\sigma^2 + \frac{1-\rho}{B}\sigma^2$, a smaller ρ lowers the irreducible floor $\rho\sigma^2$ — the part of the variance that averaging alone (bagging) can never remove.

Problem 4 — solution (c): boosting parameters

Solution P4(c) — the three tuning knobs and slow learning

`GradientBoostingRegressor(n_estimators=5000, learning_rate=0.01, max_depth=1)`:

- `n_estimators` $= B$ (*number of trees*): small trees added *sequentially*; unlike bagging, too large B *can* overfit $\Rightarrow$ choose by CV (here 5000).
- `learning_rate` $= \lambda$ (*shrinkage*, here 0.01): scales each new tree's contribution; a smaller λ learns slower and needs a larger B.
- `max_depth` $= d$ (*splits per tree / interaction depth*, here 1): the complexity of each tree; stumps ($d = 1$) give a purely additive model.

Slow learning. Each tree is fitted to the current *residuals*, and only the fraction λ of it is added. A small `learning_rate=0.01` with a large `n_estimators=5000` makes the model improve gradually, precisely where it still errs, instead of fitting hard in one step — which generalises better.

Problem 4 — solution (d): out-of-bag error

Solution P4(d) — what OOB estimates and why it is free

Out-of-bag observations. Each bootstrap sample omits on average about 1/3 of the observations (cf. P7(a): the excluded fraction $\to e^{-1} \approx 0.368$). For a fixed observation, the trees whose bootstrap sample *missed* it never saw it in training.

OOB error. Predict each observation using *only* those trees, aggregate, and compare with the truth. The resulting error is a valid estimate of the **test error** of the bagged model.

Why free. It reuses the bootstrap's own by-product — no separate validation set and no cross-validation loop are needed. (For large B it is essentially equivalent to LOOCV.)

Problem 5 (20 P) — Neural networks by hand

Lecture slides: Chapter 10 — Deep Learning (Lecture 11).

Input $x = (2{,}3)$; hidden ReLU units $g(z) = \max\{0, z\}$; linear output:

$$A_1 = g(-3 + 2x_1 + 1x_2) \qquad A_2 = g(2 - 2x_1 - 1x_2) \qquad f(x) = -2 + 3A_1 + 2A_2$$

(a) **[8 P]** Forward pass step by step: pre-activations; A_1 and A_2; output $f(x)$. What did the ReLU do to unit 2 and why are non-linear activations essential?

(b) **[6 P]** Logits $z = (2.0, 0.0, -1.0)$; true class 2: all three softmax probabilities and the cross-entropy loss ($e^2 = 7.389$; $e^0 = 1.000$; $e^{-1} = 0.368$; $\ln 8.757 = 2.170$).

(c) **[4 P]** Parameter count of a $p = 6 \to k = 5 \to K = 3$ network with biases (show both layers).

(d) **[2 P]** Conv layer: 64×64 input; $F = 3$; $P = 1$; $S = 2$: output size via $\lfloor (W - F + 2P)/S \rfloor + 1$.

Problem 5 — solution (a): forward pass

Solution P5(a) — pre-activations / activations / output

Input $x = (x_1, x_2) = (2, 3)$. **Pre-activations:**

$$z_1 = -3 + 2 \cdot 2 + 1 \cdot 3 = -3 + 4 + 3 = 4, \qquad z_2 = 2 - 2 \cdot 2 - 1 \cdot 3 = 2 - 4 - 3 = -5.$$

ReLU activations $g(z) = \max\{0, z\}$:

$$A_1 = g(4) = \max\{0, 4\} = 4, \qquad A_2 = g(-5) = \max\{0, -5\} = 0.$$

Output (linear unit):

$$f(x) = -2 + 3A_1 + 2A_2 = -2 + 3 \cdot 4 + 2 \cdot 0 = -2 + 12 + 0 = 10.$$

Comment. The ReLU *switched unit 2 off* (its negative z_2 is clipped to 0), so it contributes nothing here though it may for other inputs. Non-linearity is essential: with a linear g the network collapses into a single linear function of x regardless of depth — the ReLU is what lets hidden layers build flexible piecewise-linear surfaces.

Problem 5 — solution (b): softmax and cross-entropy

Solution P5(b) — probabilities and loss

Logits $z = (2.0, 0.0, -1.0)$. **Exponentiate and sum** (given $e^2 = 7.389$, $e^0 = 1.000$, $e^{-1} = 0.368$):

$$e^{2.0} = 7.389, \quad e^{0.0} = 1.000, \quad e^{-1.0} = 0.368 \implies \textstyle\sum_j e^{z_j} = 7.389 + 1.000 + 0.368 = 8.757.$$

Softmax $\hat{p}_k = e^{z_k} / \sum_j e^{z_j}$:

$$\hat{p}_1 = \tfrac{7.389}{8.757} = 0.844, \quad \hat{p}_2 = \tfrac{1.000}{8.757} = 0.114, \quad \hat{p}_3 = \tfrac{0.368}{8.757} = 0.042$$

(sum $= 1.000$ ✓). **Cross-entropy** at the true class 2 (given $\ln 8.757 = 2.170$):

$$-\ln \hat{p}_2 = -\ln \tfrac{1.000}{8.757} = -(0 - \ln 8.757) = \ln 8.757 = 2.170.$$

The most probable class is 1 (0.844), but the loss is scored at the *true* class 2 (0.114) — hence the sizeable loss; $\hat{p}_2 = 1$ would give loss 0.

Problem 5 — solution (c): parameter count

Solution P5(c) — parameters of a $6 \to 5 \to 3$ network

Each fully connected layer contributes (inputs $\times$ units) weights, plus one bias per unit:

Layer	Weights	Biases	Parameters
input $\to$ hidden $(6 \to 5)$	$6 \cdot 5 = 30$	5	$6 \cdot 5 + 5 = 35$
hidden $\to$ output $(5 \to 3)$	$5 \cdot 3 = 15$	3	$5 \cdot 3 + 3 = 18$
total			$35 + 18 = 53$

Don't forget the biases

A layer mapping $a \to b$ units has ab weights *and* b biases; dropping the $5 + 3 = 8$ biases would wrongly give 45.

Problem 5 — solution (d): convolution output size

Solution P5(d) — output feature-map size

Formula (per spatial dimension), with $W = 64$, $F = 3$, $P = 1$, $S = 2$:

$$\left\lfloor \frac{W - F + 2P}{S} \right\rfloor + 1 = \left\lfloor \frac{64 - 3 + 2 \cdot 1}{2} \right\rfloor + 1 = \left\lfloor \frac{63}{2} \right\rfloor + 1 = \lfloor 31.5 \rfloor + 1 = 31 + 1 = 32.$$

The same holds in both dimensions $\Rightarrow$ output feature map 32×32.

The stride $S = 2$ roughly halves the 64×64 input; the *floor* discards the fractional part ($31.5 \to 31$) *before* the $+1$ is added.

Problem 6 — task

Problem 6 (20 P) — Multiple testing

Lecture slides: Chapter 13 — Multiple Testing (Lecture 12).

(a) **[4 P]** $m = 40$ independent tests at $\alpha = 0.05$; all nulls true: define and compute the FWER ($(0.95)^{40} = 0.1285$); interpret.

(b) **[7 P]** Six ordered p-values 0.001; 0.008; 0.011; 0.030; 0.041; 0.260 — FWER control at $\alpha = 0.05$ with (i) Bonferroni and (ii) Holm; thresholds; rejected hypotheses; why is Holm never weaker than Bonferroni?

(c) **[6 P]** Benjamini–Hochberg at $q = 0.05$ on the same six p-values (0.001; 0.008; 0.011; 0.030; 0.041; 0.260): thresholds jq/m; rejections; what exactly does FDR control guarantee?

(d) **[3 P]** Confirmatory phase-III trial with $m = 5$ pre-specified endpoints (one false positive $\Rightarrow$ an ineffective drug approved): FWER or FDR? Justify.

Problem 6 — solution (a): FWER

Solution P6(a) — FWER for 40 independent tests

Definition. With V = number of true nulls that are falsely rejected,

$$\mathrm{FWER} = \Pr(V \geq 1) = \Pr(\text{at least one Type I error}).$$

Computation. All 40 nulls true, tests independent, each at level 0.05:

$$\mathrm{FWER} = 1 - \Pr(\text{no false rejection}) = 1 - (1 - 0.05)^{40} = 1 - (0.95)^{40} = 1 - 0.1285 = 0.8715.$$

Interpretation. Although each single test errs only 5% of the time, the chance of *at least one* spurious "discovery" among 40 unadjusted tests is about 87% — multiple testing without adjustment almost guarantees false findings.

Problem 6 — solution (b): Bonferroni

Solution P6(b) part 1 — Bonferroni at $\alpha = 0.05$

Rule. Reject $H_{(j)}$ iff $p_{(j)} \leq \dfrac{\alpha}{m} = \dfrac{0.05}{6} = 0.00833$ — the *same* threshold for every j.

j	1	2	3	4	5	6
$p_{(j)}$	0.001	0.008	0.011	0.030	0.041	0.260
≤ 0.00833?	✓	✓	×	×	×	×

$p_{(1)} = 0.001$ and $p_{(2)} = 0.008$ pass; $p_{(3)} = 0.011 > 0.00833$ fails.

$$\Longrightarrow \text{ reject } H_{(1)}, H_{(2)} : \ \textbf{2 rejections}.$$

Problem 6 — solution (b): Holm step-down

Solution P6(b) part 2 — Holm and why Holm $\supseteq$ Bonferroni

Rule. Compare $p_{(j)}$ with $\dfrac{\alpha}{m-j+1}$ in order; *stop at the first failure* and reject everything before it.

j	1	2	3	4	5	6
$p_{(j)}$	0.001	0.008	0.011	0.030	0.041	0.260
$\alpha/(m-j+1)$	0.00833	0.0100	0.0125	0.0167	0.0250	0.0500
decision	✓	✓	✓	**stop**	—	—

$0.001 \le 0.00833$, $0.008 \le 0.0100$, $0.011 \le 0.0125$, but $0.030 > 0.0167 \Rightarrow$ stop: reject $H_{(1)}, H_{(2)}, H_{(3)}$ — **3 rejections**.

Holm $\supseteq$ Bonferroni. The hurdles satisfy $\dfrac{\alpha}{m-j+1} \ge \dfrac{\alpha}{m}$ for all j (equality only at $j=1$), so Holm is uniformly more powerful yet still controls FWER without any independence assumption. Here it rejects one more ($H_{(3)}$).

Problem 6 — solution (c): Benjamini–Hochberg

Solution P6(c) — BH at FDR level $q = 0.05$

Rule. Find the largest j with $p_{(j)} \le \dfrac{jq}{m} = \dfrac{j \cdot 0.05}{6}$, then reject $H_{(1)}, \ldots, H_{(L)}$.

j	1	2	3	4	5	6
$p_{(j)}$	0.001	0.008	0.011	0.030	0.041	0.260
jq/m	0.00833	0.0167	0.0250	0.0333	0.0417	0.0500
$p_{(j)} \le jq/m$?	yes	yes	yes	yes	**yes**	no

Largest passing index $L = 5$ ($0.041 \le 0.0417$; $0.260 > 0.0500$) $\Rightarrow$ reject $H_{(1)}, \ldots, H_{(5)}$ — **5 rejections** (most liberal: 5 vs. Holm 3, Bonferroni 2).

Guarantee. $\text{FDR} = E\left[\dfrac{V}{\max(R,1)}\right] \le q = 0.05$: *on average* at most 5% of rejections are false. It does *not* bound $\Pr(V \ge 1)$ (that is FWER), nor guarantee $\le 5\%$ false among these particular 5.

Problem 6 — solution (d): FWER or FDR?

Solution P6(d) — confirmatory phase-III trial

Recommendation: control the FWER.

Why. This is a *confirmatory* study with only $m = 5$ pre-specified endpoints, and a single false positive is catastrophic (an ineffective drug approved; patient harm; regulatory failure). FWER control (e.g. Holm at $\alpha = 0.05$) bounds the probability of *any* false rejection at 5% — exactly the guarantee a regulator demands.

Why not FDR. FDR only limits the *expected proportion* of false endpoints and would tolerate an occasional false positive. That is fine for large exploratory screens, but not when each individual claim must stand on its own.

Problem 7 (20 P) — Cumulative essentials

Lecture slides: Chapters 2–6 (Lectures 1–8).

(a) **[4 P]** Bootstrap: (i) show Pr(obs. not in sample) $= (1 - 1/n)^n$ and its limit ($e^{-1} = 0.368$); (ii) fraction included as $n \to \infty$; (iii) exact inclusion probability for $n = 5$ ($0.8^5 = 0.328$).

(b) **[4 P]** Bias–variance at x_0 read from a pandas `DataFrame` — `flexible_spline`: `var`$= 3.5$, `bias`$= -1.0$, `noise_var`$= 2.0$; `linear_model`: `var`$= 0.5$, `bias`$= -2.5$, `noise_var`$= 2.0$. Expected test MSE of each; which to deploy; lowest achievable MSE?

(c) **[3 P]** Elastic-net penalty: what does it inherit from lasso and from ridge? When does it clearly beat the lasso?

(d) **[3 P]** Collinearity (corr. 0.95): effect on OLS estimates; $\mathrm{VIF}_j = 1/(1 - R_j^2)$ for $R_j^2 = 0.9$ ($\sqrt{10} = 3.162$).

(e) **[6 P]** Match to {linear regression; lasso; logistic regression; random forest}, each once: (i) Δwage per school-year with CI ($n = 3{,}000$); (ii) biomarker from 8,000 metabolites ($n = 80$) with a short list; (iii) churn yes/no with odds for management; (iv) electricity demand from hundreds of interacting variables, accuracy only.

Problem 7 — solution (a): bootstrap inclusion

Solution P7(a) — inclusion probability

(i) In each of the n independent draws a fixed observation is *missed* with probability $1 - \frac{1}{n}$; all n draws miss it with probability

$$\left(1 - \frac{1}{n}\right)^n \xrightarrow[n\to\infty]{} e^{-1} = 0.368.$$

(ii) Hence it is *included* with probability $1 - e^{-1} = 1 - 0.368 = 0.632$: on average about 63.2% of observations appear in a bootstrap sample (the other $\approx 1/3$ are the out-of-bag observations).

(iii) For $n = 5$:

$$\Pr(\text{included}) = 1 - \left(1 - \frac{1}{5}\right)^5 = 1 - 0.8^5 = 1 - 0.328 = 0.672.$$

Problem 7 — solution (b): bias–variance decomposition

Solution P7(b) — expected test MSE

Decomposition: $\mathrm{MSE} = \mathrm{Var} + \mathrm{Bias}^2 + \mathrm{Var}(\varepsilon)$. Reading each row of the `DataFrame`:

$$\mathrm{MSE}_{\texttt{flexible\_spline}} = 3.5 + (-1.0)^2 + 2.0 = 3.5 + 1.0 + 2.0 = 6.5,$$

$$\mathrm{MSE}_{\texttt{linear\_model}} = 0.5 + (-2.5)^2 + 2.0 = 0.5 + 6.25 + 2.0 = 8.75.$$

Deploy `flexible_spline` ($6.5 < 8.75$): its higher variance (3.5 vs. 0.5) is more than offset by its much smaller squared bias (1.0 vs. 6.25).

Lowest achievable MSE = the irreducible error $\mathrm{Var}(\varepsilon) =$ `noise_var` $= 2.0$; no method can go below it.

Problem 7 — solution (c): elastic net

Solution P7(c) — elastic-net penalty

Penalty (a mix of ℓ_2 and ℓ_1, for $0 \leq \alpha \leq 1$):

$$\lambda\Big[\tfrac{1-\alpha}{2}\sum_j \beta_j^2 + \alpha\sum_j |\beta_j|\Big].$$

From the lasso (ℓ_1): exact zeros $\Rightarrow$ variable selection / sparsity.
From ridge (ℓ_2): graceful handling of correlated predictors, and the ability to select more than n variables when $p > n$.

When it clearly beats the lasso: with *groups of highly correlated predictors*. The lasso tends to keep one member of the group and drop the rest (essentially at random); the elastic net's ridge part shares the weight across the group and keeps them in or out together — the *grouping effect*.

Problem 7 — solution (d): collinearity and VIF

Solution P7(d) — effect on OLS and the VIF

Effect on OLS. Collinearity does *not* bias the estimates, but makes them *imprecise and unstable*: it inflates variances and standard errors, widens confidence intervals, and can make coefficients flip sign or lose significance (predictions may still be fine).

VIF for $R_j^2 = 0.9$:

$$\mathrm{VIF}_j = \frac{1}{1 - R_j^2} = \frac{1}{1 - 0.9} = \frac{1}{0.1} = 10.$$

So $\mathrm{Var}(\hat{\beta}_j)$ is $10\times$ larger than with an uncorrelated X_j, and its standard error is inflated by $\sqrt{10} = 3.162$. Rule of thumb: $\mathrm{VIF} > 5$–10 signals problematic collinearity.

Problem 7 — solution (e): matching methods to scenarios

Solution P7(e) — each method used exactly once

(i) **Linear regression** — quantitative response, goal is *inference* (effect size + CI), $n = 3{,}000 \gg p$: OLS with standard errors delivers exactly this.

(ii) **Lasso** — $p = 8{,}000 \gg n = 80$ makes OLS infeasible; the ℓ_1 penalty regularises and returns a short list of relevant metabolites.

(iii) **Logistic regression** — binary response; coefficients give odds ratios $e^{\hat{\beta}_j}$, ideal for reporting how each characteristic changes the odds.

(iv) **Random forest** — captures non-linearities and interactions among hundreds of predictors, accurate and robust; interpretability is not required, only forecast accuracy.